

ASSIGNMENT-1

Design and implement a Java program that performs arithmetic operations using a menu-driven approach with proper user input validation and modular methods.

CODE:

```
import java.util.Scanner;

public class Calculator {

    public static int addnums(int n1, int n2) {
        return n1 + n2;
    }

    public static int subnums(int n1, int n2) {
        return n1 - n2;
    }

    public static long mulnums(int n1, int n2) {
        return (long) n1 * n2;
    }

    public static float divnums(int n1, int n2) {
        return (float) n1 / n2;
    }

    public static int modulus(int n1, int n2) {
        return n1 % n2;
    }
}
```

```
}
```

```
public static void main(String[] args) {
```

```
    Scanner scan = new Scanner(System.in);
```

```
    int opr;
```

```
    do {
```

```
        System.out.println("\n--- Calculator Menu ---");
```

```
        System.out.println("0. Exit");
```

```
        System.out.println("1. Addition");
```

```
        System.out.println("2. Subtraction");
```

```
        System.out.println("3. Multiplication");
```

```
        System.out.println("4. Division");
```

```
        System.out.println("5. Modulus");
```

```
        System.out.print("Enter your choice: ");
```

```
        opr = scan.nextInt();
```

```
        if (opr == 0) {
```

```
            System.out.println("Exiting Calculator...");
```

```
            break;
```

```
        }
```

```
        System.out.print("Enter first number: ");
```

```
        int n1 = scan.nextInt();
```

```
        System.out.print("Enter second number: ");
```

```
int n2 = scan.nextInt();
```

```
switch (opr) {
```

```
    case 1:
```

```
        System.out.println("Sum = " + addnums(n1, n2));
```

```
        break;
```

```
    case 2:
```

```
        System.out.println("Difference = " + subnums(n1, n2));
```

```
        break;
```

```
    case 3:
```

```
        System.out.println("Product = " + mulnums(n1, n2));
```

```
        break;
```

```
    case 4:
```

```
        System.out.println("Quotient = " + divnums(n1, n2));
```

```
        break;
```

```
    case 5:
```

```
        System.out.println("Remainder = " + modulus(n1, n2));
```

```
        break;
```

```
    default:
```

```
        System.out.println("Invalid choice!");
```

```
}
```

```
} while (true);
```

```
scan.close();
```

```
}
```

```
}
```

OUTPUT:

(a) Addition

```
--- Calculator Menu ---
0. Exit
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Modulus
Enter your choice: 1
Enter first number: 25
Enter second number: 5
Sum = 30
```

(b) Subtraction

```
--- Calculator Menu ---
0. Exit
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Modulus
Enter your choice: 2
Enter first number: 52
Enter second number: 45
Difference = 7
```

(c) Multiplication

```
--- Calculator Menu ---
0. Exit
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Modulus
Enter your choice: 3
Enter first number: 100
Enter second number: 965
Product = 96500
```

(d) Division

```
--- Calculator Menu ---
0. Exit
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Modulus
Enter your choice: 4
Enter first number: 85
Enter second number: 20
Quotient = 4.25
```

(e) Division with n2 as 0

```
--- Calculator Menu ---
0. Exit
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Modulus
Enter your choice: 4
Enter first number: 56
Enter second number: 0
Quotient = Infinity
```

(f) No valid input of number

```
--- Calculator Menu ---
0. Exit
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Modulus
Enter your choice: 2
Enter first number: ds
Exception in thread "main" java.util.InputMismatchException
    at java.base/java.util.Scanner.throwFor(Scanner.java:939)
    at java.base/java.util.Scanner.next(Scanner.java:1594)
    at java.base/java.util.Scanner.nextInt(Scanner.java:2258)
    at java.base/java.util.Scanner.nextInt(Scanner.java:2212)
    at Calculator.main(Calculator.java:47)
```

(g) No valid input of the operator

```
--- Calculator Menu ---  
0. Exit  
1. Addition  
2. Subtraction  
3. Multiplication  
4. Division  
5. Modulus  
Enter your choice: 7  
Enter first number: 8  
Enter second number: 3  
Invalid choice!
```

(h) Exiting

```
--- Calculator Menu ---  
0. Exit  
1. Addition  
2. Subtraction  
3. Multiplication  
4. Division  
5. Modulus  
Enter your choice: 0  
Exiting Calculator...  
PS C:\Users\Kunal_jhindal\OneDrive\Desktop\Sem4\PJL\Assignment 1>
```